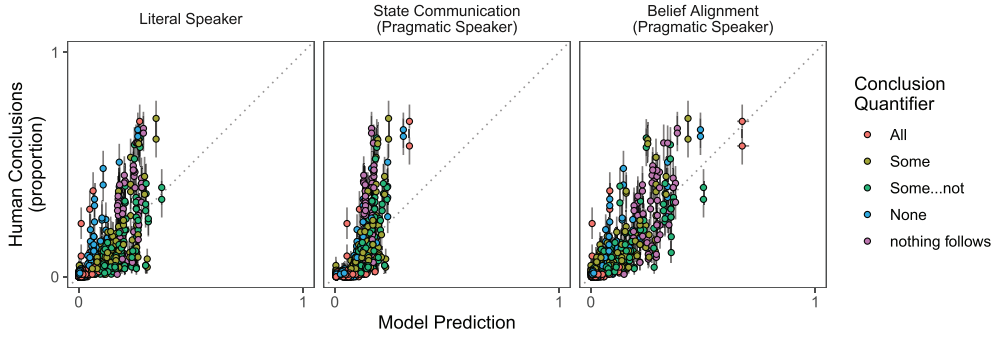


Fig. 7. Posterior predictive model fits for the literal speaker model and two versions of the pragmatic speaker model. Each point represents one of nine possible conclusions for one of the 64 possible syllogisms. Error bars denote bootstrapped 95% confidence intervals for the empirical data. Posterior model predictions did not exhibit significant variance; the horizontal error bars that are large enough to see represent the 95% Bayesian credible intervals for the model predictions.

were not appreciably different.[6] We performed model comparison by computing Bayes Factors, which quantify the likelihood of the data under each model averaging over the model's prior distribution over parameters. By averaging over each model's prior on parameters, Bayes factors implicitly penalize models with extra degrees of freedom (Lee & Wagenmakers, 2014). We estimated the marginal likelihoods of the data under each model using an Annealed Importance Sampling algorithm (Neal, 2001).

3.1.3. Results

Overall, we found that the Belief Alignment pragmatic speaker model predicted the empirical data the best and provided the most parsimonious explanation of the three models (Fig. 7). This model performed several orders of magnitude better at explaining the data, taking into the account model complexity (in comparison to the Literal Speaker model, $\log BF \approx 300$; in comparison to the State Communication pragmatic speaker $\log BF \approx 1200$). It also achieved the highest correlation coefficient and lowest mean squared error of the RSA model variants (Table 2).

The Belief Alignment pragmatic speaker model captures a number of interesting patterns in the human syllogistic reasoning behavior. By incorporating a notion of informativeness into the selection of a conclusion, the model is able to show a preference among equally true conclusions. For example, in the *All As are Bs/All Bs are Cs* syllogism, the conclusion *Some As are Cs* is true in every possible world that *All As are Cs* is true, because *all* entails *some*. Nonetheless, the Belief Alignment pragmatic speaker model and human participants strongly prefer the *all* conclusion (Fig. 8, top-left facet); the model arrives at this preference because *all* more informatively conveys the belief distribution of the speaker in comparison to *some*. Syllogisms for which the premises provide little information about the conclusion also reveal interesting reasoning patterns in both participants and the models. In the *No As are Bs/No Bs are Cs* syllogism, participants tend to respond either *nothing follows* (the deductively correct conclusion) or either *No As are Cs* or *No Cs are As* (Fig. 9, second row, second column).

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